

VLSI INTERNSHIP – TASK 1

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Domain: Very Large Scale Integration (VLSI) Intern

Goal :

Write RTL for a small digital block (e.g., 8-bit ALU or simple RISC datapath) and verify functionality with testbenches.

Requirements:

- Create synthesizable Verilog/SystemVerilog modules.
- Write comprehensive testbench with directed and random tests.
- Simulate using ModelSim/GTKWave or any simulator.

8-bit ALU Verilog code:

```
module alu8bit(  
    input clk,  
    input rst,  
    input [7:0] A, B,  
    input [2:0] sel,  
    output reg [7:0] Y  
);  
  
always @(posedge clk or posedge rst)  
    if (rst)  
        Y <= 0;  
    else  
        case(sel)  
            3'b000: Y <= A + B;  
            3'b001: Y <= A - B;  
            3'b010: Y <= A & B;  
            3'b011: Y <= A | B;  
            3'b100: Y <= A ^ B;  
            3'b101: Y <= ~A;  
            3'b110: Y <= A << 1;  
            3'b111: Y <= A >> 1;  
        endcase  
    endmodule
```

Testbench:

```
`timescale 1ns/1ps

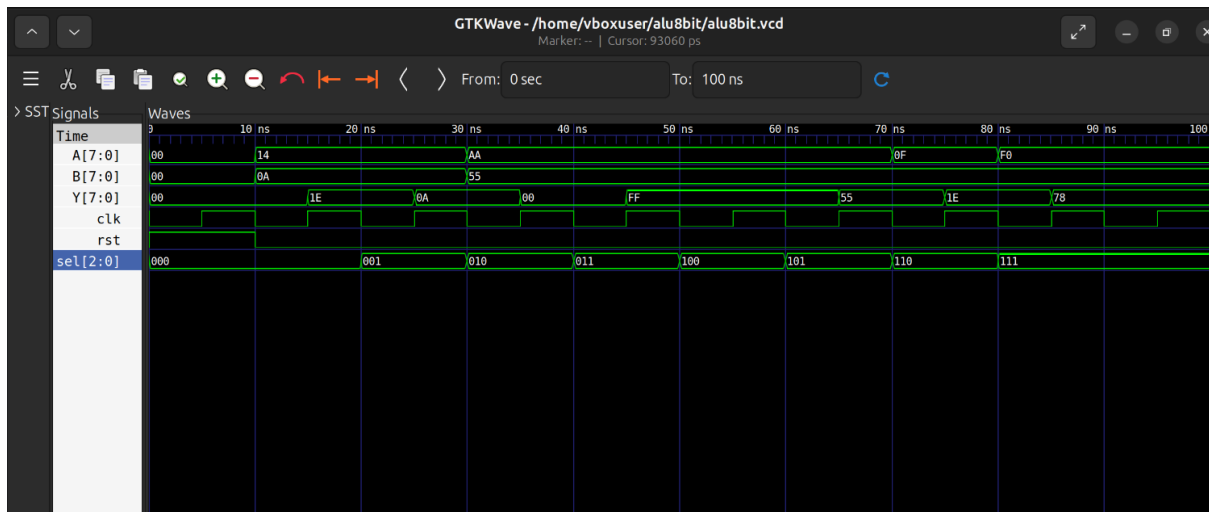
module alu8bit_tb;
reg clk;
reg rst;
reg [7:0] A, B;
reg [2:0] sel;
wire [7:0] Y;

alu8bit uut (.clk(clk),.rst(rst),.A(A),.B(B),.sel(sel),.Y(Y));
always #5 clk = ~clk;
initial begin
    $dumpfile("alu8bit.vcd");
    $dumpvars(0, alu8bit_tb);
    clk = 0;
    rst = 1;
    A = 0;
    B = 0;
    sel = 0;

    #10 rst = 0;

    A = 8'd20; B = 8'd10; sel = 3'b000; #10;
    A = 8'd20; B = 8'd10; sel = 3'b001; #10;
    A = 8'hAA; B = 8'h55; sel = 3'b010; #10;
    A = 8'hAA; B = 8'h55; sel = 3'b011; #10;
    A = 8'hAA; B = 8'h55; sel = 3'b100; #10;
    A = 8'hAA;           sel = 3'b101; #10;
    A = 8'h0F;           sel = 3'b110; #10;
    A = 8'hF0;           sel = 3'b111; #10;
    #10 $finish;
end
initial begin
    $monitor("Time=%0t rst=%b sel=%b A=%h B=%h Y=%h",
        $time, rst, sel, A, B, Y);
end
endmodule
```

GTKWave:



Commands:

```
vboxuser@ubuntu:~$ mkdir ~/alu8bit
cd ~/alu8bit
vboxuser@ubuntu:~/alu8bit$ nano alu8bit.v
vboxuser@ubuntu:~/alu8bit$ nano alu8bit.v
vboxuser@ubuntu:~/alu8bit$ nano alu8bit_tb.v
vboxuser@ubuntu:~/alu8bit$ iverilog -o alu8bit_out alu8bit.v alu8bit_tb.v
vboxuser@ubuntu:~/alu8bit$ vvp alu8bit_out
VCD info: dumpfile alu8bit.vcd opened for output.
Time=0 rst=1 sel=000 A=00 B=00 Y=00
Time=10000 rst=0 sel=000 A=14 B=0a Y=00
Time=15000 rst=0 sel=000 A=14 B=0a Y=1e
Time=20000 rst=0 sel=001 A=14 B=0a Y=1e
Time=25000 rst=0 sel=001 A=14 B=0a Y=0a
Time=30000 rst=0 sel=010 A=aa B=55 Y=0a
Time=35000 rst=0 sel=010 A=aa B=55 Y=00
Time=40000 rst=0 sel=011 A=aa B=55 Y=00
Time=45000 rst=0 sel=011 A=aa B=55 Y=ff
Time=50000 rst=0 sel=100 A=aa B=55 Y=ff
Time=60000 rst=0 sel=101 A=aa B=55 Y=ff
Time=65000 rst=0 sel=101 A=aa B=55 Y=55
Time=70000 rst=0 sel=110 A=0f B=55 Y=55
Time=75000 rst=0 sel=110 A=0f B=55 Y=1e
Time=80000 rst=0 sel=111 A=f0 B=55 Y=1e
Time=85000 rst=0 sel=111 A=f0 B=55 Y=78
alu8bit_tb.v:44: $finish called at 100000 (1ps)
```